

How simple the matrix can be made

Choosing a basis means choosing P , and the question is how simple $P^{-1}AP$ can be made. The answer depends entirely on what A satisfies — and the list is ordered from the strongest structure to no structure at all.

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DIAGONAL — THE BEST CASE

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1

Distinct eigenvalues

n distinct λ

i

$$D = P^{-1}AP$$

Eigenvectors for distinct eigenvalues are automatically independent, so P is invertible without further checking. Sufficient but not necessary — the row below covers the repeated case, and I diagonalizes trivially with one eigenvalue.

2

Multiplicities agree

$m(\lambda_i) = m(A - \lambda_i I)$ for every i

$g_i = a_i$

$$D = P^{-1}AP$$

The **necessary and sufficient condition**. Columns of P are n independent eigenvectors; the question is only whether that many exist, and the multiplicity comparison answers it eigenvalue by eigenvalue.

3

Real symmetric

$A = A^T$ — always succeeds

T

$$D = P^{-1}AP$$

The **spectral theorem** guarantees not just diagonalizability but an orthonormal eigenbasis, so $P^{-1} = P^T$ and nothing needs inverting. The only row where the hypothesis is checkable at a glance.

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AS CLOSE AS THE MATRIX ALLOWS

2

4

Defective

$m(\lambda) < m(A - \lambda I)$ for some λ

$g < a$

$$J = P^{-1}AP$$

Block diagonal with ones on the superdiagonal. The columns of P are eigenvectors plus generalized eigenvectors, and the number of ones counts exactly how many eigenvectors are missing — the Jordan form measures the failure rather than working around it.

5

Any complex square matrix

no hypotheses; U unitary

$$T = U^{-1}AU, \text{ upper triangular}$$

Schur decomposition. Triangular rather than diagonal, but it exists for every square matrix over $\mathbb{C}$ and U is unitary, which makes it numerically stable where the Jordan form is not — the reason libraries compute Schur and not Jordan.

Every row is the same operation with a different guarantee: conjugate by P and read the result. Similarity preserves eigenvalues, trace, determinant, rank and characteristic polynomial, so nothing about the transformation is lost in any of these — only the coordinates change, and the entries below say how much simplicity that buys.